SEBASTIAN MOCANU

Research Scientist - PhD Candidate, Computer Vision & Robotics

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SUMMARY

PhD candidate at National University of Science and Technology POLITEHNICA Bucharest (UNSTPB) researching vision-based control and perception for autonomous drones, focusing on self-supervised learning, sim-to-real transfer and efficient real-time neural networks. Building systems end to end, from Unreal Engine digital twins to models flying on real quadrotors.

EXPERIENCE

Research Scientist

UNSTPB

Feb 2024 – Present Bucharest, Romania

- Researcher on **DIGITWINS4CIUE** (Digital Europe Program) and, since May 2025, **HRIA**, Romania's national AI research hub.
- Built an Unreal Engine digital twin of the UNSTPB PRECIS laboratory, enabling real-world test replication and validation in a simulation environment.
- Conducting research on vision-based control and perception for drones such as self-supervised visual servoing (ICCVW 2025, oral), vision-only indoor navigation (under review), and continual learning for object perception (ERF 2026), validated in both simulation and real hardware.
- Contributing to project reporting and coordination. Co-author of the DIGITWINS4CIUE white paper.
- *Stack:* Python, PyTorch, C++, OpenCV, Unreal Engine, Linux, Git, Parrot Sphinx, ROS, Gazebo.

Scientific Research Assistant - UNSTPB & Google

UNSTPB

Jun 2022 – Nov 2023 Bucharest, Romania

- Selected for a Google-funded research scholarship based on academic results. Part-time from Jun 2022, full-time from Apr 2023.
- Built a quadrotor that flies autonomously from a single camera, combining computer vision, control and reinforcement learning to obtain continuous in-scene flight, obstacle avoidance, safe-landing zone detection and gesture-command control and person following.
- *Stack:* Python, PyTorch, C++, OpenCV, Linux, Git.

Software Developer

Endava

Dec 2021 – Apr 2023 Bucharest, Romania

- Worked in an Agile team, on the core legacy payment-processing platform of a Tier-1 global bank, migrating critical parts of the data layer from Oracle to MongoDB speeding up reads and response time.

EDUCATION

PhD in Computer Science - AI

Doctoral School of Engineering and Applications of Lasers and Accelerators

Oct 2023 – Present UNSTPB

Advisors: Prof. Dr. Marius Leordeanu, Prof. Dr. Ing. Emil Slușanschi. Visual servoing, vision-based UAV flight, continual learning for drones.

MSc in Artificial Intelligence

Faculty of Automatic Control and Computers

Oct 2021 – Sep 2023 UNSTPB

Coordinator: Prof. Dr. Marius Leordeanu. Thesis on vision-based autonomous drone control. Teaching diploma Psycho-Pedagogical Module II.

BEng Mechatronics & Robotics

Faculty of Mechanical Engineering and Mechatronics

Sep 2017 – Jul 2021 UNSTPB

Built a drone and a robotic arm. Teaching diploma Psycho-Pedagogical Module I.

AWARDS

1st place, CLEF 2026 JOKER
Task 1, humor retrieval (English)

2nd place, UPB Student Scientific Session 2023
Scene Surveillance and Mapping for Self-Flying Drones

3rd place, UPB Student Scientific Session 2022
From the Mind to the Sky: Teaching Drones How to Fly

PROJECTS

NeuralFactory: Real-Time Robot Simulation in Unreal Engine 5

- UE5 C++ robot that streams its camera feed to a Python controller over WebSocket, forming a closed perception-action loop at 30 Hz.
- Async GPU readback and off-thread JPEG encoding to avoid game-thread stalls which uses a custom binary protocol with ack-based backpressure, telemetry and auto-reconnect.
- *Stack: C++, Unreal Engine 5, Python, asyncio, OpenCV, WebSockets*

- On my own initiative, built a full-stack Amazon S3 browser (Java/SpringBoot, JavaScript) that my 10-person team adopted in place of a paid per-seat viewer application license. It was later added to other teams repositories.
- *Stack:* Java, Spring Boot, Maven, AWS, Amazon S3, HTML, CSS, JavaScript, Oracle Database, MongoDB, Git, Bash, Linux.

PUBLICATIONS

Full list on Google Scholar. Code, models and datasets on GitHub and Hugging Face.

- **S. Mocanu**, S.-I. Nae, M.-E. Barbu, M. Leordeanu. *Efficient Self-Supervised Neuro-Analytic Visual Servoing for Real-time Quadrotor Control*. ICCV 2025 Workshops. **Oral. Poster.** [paper] [code] [website]
- **S. Mocanu**, E. Slusanschi, M. Leordeanu. *Self-Supervised Learning to Fly using Efficient Semantic Segmentation and Metric Depth Estimation for Low-Cost Autonomous UAVs*. Under review, 2025. [arXiv]
- S.-I. Nae, M.-E. Barbu, **S. Mocanu**, M. Leordeanu. *Learning on the Fly: Replay-Based Continual Object Perception for Indoor Drones*. European Robotics Forum (ERF) 2026. **Oral. Poster.** [paper] [website]
- A.-M. L. Mocanu, **S. Mocanu**, C.-O. Truică, E.-S. Apostol. *Through the Eyes of the Beholder: Biometric and Demographic Conditioning for Multimodal Sexism Detection*. CLEF 2026, EXIST Lab. **Oral. Poster.** [paper] [code] [website]
- A.-M. L. Mocanu, **S. Mocanu**, C.-O. Truică, E.-S. Apostol. *IROH: Insightful Ranking Of Humor using Multi-Stage Hybrid Retrieval with Rationale-Distilled LLM Judges*. CLEF 2026, JOKER Lab. **1st place, Task 1 (English). Oral.** [paper] [code] [website]

SKILLS

Research

Self-Supervised Learning, Knowledge Distillation, Visual Servoing, Semantic Segmentation, Monocular Depth Estimation, Reinforcement Learning, Continual Learning, LLM fine-tuning (LoRA) and retrieval, Vision-Language Models.

Machine Learning Stack

PyTorch | OpenCV | CUDA | W&B | Hugging Face

Robotics & Simulation

Unreal Engine | ROS | Gazebo | Parrot Sphinx | Arduino | Raspberry Pi

Programming Languages

Python | C++ | Java | C | MATLAB | SQL | JavaScript

Engineering

Linux | Git | Docker | CMake | CI/CD | Spring Boot | Django | MongoDB | Oracle | SQLite | AWS

Semantic Search for Images, Videos and GIFs

- Full-stack app for searching a personal media library by text prompt or by example image, video or GIF, with a results view similar to Google Images or YouTube.
- Python package that finds similar media from an image or text query using CLIP embeddings, with frame-sampling strategies for videos and GIFs.
- PostgreSQL store for embeddings with multiple CLIP models, with incremental re-indexing of changed files and batched processing for large collections.
- *Stack: Python, CLIP (sentence-transformers), PostgreSQL, OpenCV, Flask, JavaScript*

Abusive Language Detection in Social Media - Natural Language Processing

- Benchmarked 7 deep learning models on 5 hate-speech datasets (35 total models) plus ensembles, with per-dataset analysis and preprocessing.
- Performed tests on HateCheck to evaluate each model and ensemble. [code]

RL Crafter - Reinforcement Learning

- Implemented and compared DQN, Double DQN, Dueling DQN, Dueling Double DQN and C51 on the Crafter survival benchmark over 3 seeds. Dueling DQN performed the best. [code]

TMETF Department Website UPB

- Led the development of a comprehensive website from its inception, specifically tailored to cater to the information needs of students enrolled in the Thermal Systems and Equipment (SET) program. This project delivers a user-friendly and informative platform, designed to aid administrators with robust control and flexibility.
- Architecture was crafted to ensure effortless management, giving the administrators the capability to modify every page without code. http://termo.pub.ro/

HOBBIES

- Guitar and violin since age 12, with a 3-year arts school diploma in guitar and music theory.
- Game development in Unreal Engine and Godot.
- Electronics and robotics prototyping with Arduino and Raspberry Pi.
- Building tools for my cat, Maple, such as a web app that tracks her water intake. [code]